Destination Earth - Data discovery and Management Tool

Release 1.0

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ONE

SRC

1.1 masterStructure module

This module contains the core structural elements of the data discovery tool. The core structural elements consists of the names of applications, HPC centers, ESMs etc. The names of the file types produced by each application and their extensions. Apart from that certain data structures to hold these infos.

`masterStructure.appDataPath = '/work/bm0146/k204247/DestinE/destinE_DDTool/destinEData'`

Path to location of the data produced by the applications.

Type
str

`masterStructure.appDataSrcNameFileExt = {'image': ['*.jpg', '*.pdf', '*.png', '*.gif'], 'netcdf': ['*.nc'], 'text': ['*.txt', '*.csv']}`

List of the file extensions.

Type
list[str]

`masterStructure.appDataSrcNames = ['netcdf', 'image', 'text']`

List of the file types.

Type
list[str]

`masterStructure.appNamesList = ['AQUA', 'EnergyOnShore', 'EnergyOffShore', 'FWI', 'HydroMet', 'HydroRiver', 'SPITFIRE', 'Urban', 'WISE']`

List of the application names.

Type
list[str]

`masterStructure.appSrcFlags = {'AQUA': [True, True, False], 'EnergyOffShore': [True, True, False], 'EnergyOnShore': [True, True, True], 'FWI': [True, True, False], 'HydroMet': [True, False, True], 'HydroRiver': [True, False, True], 'SPITFIRE': [True, False, False], 'Urban': [True, True, True], 'WISE': [False, True, False]}`

Dictionary of the file types produced by all the apps.

Type
dict[str]

`masterStructure.ddtBasePath = '/work/bm0146/k204247/DestinE/destinE_DDTool'`

Path to location of the data discovery and management tool on disk.

Type
str

`masterStructure.esmNames = ['ICON', 'IFS']`

List of the ESMs.

Type
list[str]

`masterStructure.hpcCenters = ['LUMI', 'MareNostrum']`

List of the HPC centers.

Type
list[str]

1.2 AQUA module

This module contains the information about the AQUA app.

1.3 EnergyOffShore module

This module contains the information about the EnergyOffShore app.

1.4 EnergyOnShore module

This module contains the information about the EnergyOnShore app.

1.5 FWI module

This module contains the information about the FWI app.

1.6 HydroMet module

This module contains the information about the HydroMet app.

1.7 HydroRiver module

This module contains the information about the HydroRiver app.

1.8 SPITFIRE module

This module contains the information about the SPITFIRE app.

1.9 Urban module

This module contains the information about the Urban app.

1.10 WISE module

This module contains the information about the WISE app.

1.11 utils module

This module contains certain utitlity functions like for fetching the files from a given location on disk etc.

`utils.`**`getAppSrcFileList`**`(`*app*, *src*, *srcExtList*`)`

Create a list of files produced by the app, for a given file type.

Search the app data directory and create a list of all the files of a given file type like netcdf, jpeg etc.

Parameters

- **`app`** – Application Name.
- **`src`** – Source or file type name like netcdf, image, text.
- **`srcExt`** – The file suffix like *.nc, *.jpg, *.pdf, *.txt, *.csv.

Returns

srcFileList - a list.

1.12 intakeCuration module

This module contains the APIs for updating the intake catalog for each app and each of it's file types. These APIs need to be called by the app data curator for maintaining the intake catalog.

`intakeCuration.`**`createAppDataSrcNamesIntakeAPImap`**`()`

Create a mapping between the file types and the associated API for handling it.

Mapper to club the file types and the corresponding access methods for handling netcdf, jpeg etc.

Parameters

`None.` –

Returns

None

`intakeCuration.`**`createAppIntakeCatalog`**`()`

Create yaml files for the Apps.

For each of the Apps, create the yaml files containing the links to the yaml files for the ESMs .

Parameters
`None.` –

Returns
None

`intakeCuration.createDDTMasterIntakeCatalog()`

Create master catalog file for the HPC centers.

Create the top level master catalog containing the links to the yaml files for the HPC centers.

Parameters
`None.` –

Returns
None

`intakeCuration.createESMIntakeCatalog()`

Create yaml files for the ESMs.

For each of the ESMs, create the yaml files containing the links to the yaml files for the available sources .

Parameters
`None.` –

Returns
None

`intakeCuration.createHPCIntakeCatalog()`

Create yaml files for the HPC centers.

For each of the HPC centers, create the yaml files containing the links to the yaml files for the Apps .

Parameters
`None.` –

Returns
None

`intakeCuration.createImageSrcListForIntake(`*catFile*, *srcFileList*`)`

Create yaml sources for the image files in the input
file list of files.

For each of the image files in the input file list, create the yaml source and save in the input catalog file.

Parameters

- **`catFile`** – Catalog File Name.
- **`srcFileList`** – List of image files.

Returns
None

`intakeCuration.createNetcdfSrcListForIntake(`*catFile*, *srcFileList*`)`

Create yaml sources for the netcdf files in the input
file list of files.

For each of the netcdf files in the input file list, create the yaml source and save in the input catalog file.

Parameters

- **`catFile`** – Catalog File Name.

- **srcFileList** – List of netcdf files.

Returns
None

`intakeCuration.createSrcIntakeCatalog()`

Create yaml files for the sources.

For each of the app, check the available sources (file types) and create the corresponding yaml files.

Parameters
None. –

Returns
None

`intakeCuration.createTextSrcListForIntake(catFile, srcFileList)`

Create yaml sources for the text files in the input
file list of files.

For each of the text files in the input file list, create the yaml source and save in the input catalog file.

Parameters

- **catFile** – Catalog File Name.
- **srcFileList** – List of text files.

Returns
None

1.13 stacCuration module

This module contains the APIs for updating the STAC catalog for each app and each of it's file types. These APIs need to be called by the app data curator for maintaining the STAC catalog.

`stacCuration.createAppDataSrcNamesSTACAPImap()`

Create a mapping between the file types and the associated STAC API for handling it.

Mapper to club the file types and the corresponding access methods for handling netcdf, jpeg etc.

Parameters
None. –

Returns
None

`stacCuration.createAppSTACCatalog()`

Create STAC catalog for the Apps.

For each of the Apps, create the STAC catalog containing the links to the STAC catalogs for the ESMs.

Parameters
None. –

Returns
None.

`stacCuration.`**`createDDTMasterSTACCatalog`**`()`

Create master STAC catalog file for the HPC centers.

Create the top level master STAC catalog containing the links to the STAC catalogs for the HPC centers.

Parameters
`None.` –

Returns
None.

`stacCuration.`**`createESMSTACCatalog`**`()`

Create STAC catalog for the ESMs.

For each of the ESMs, create the STAC catalog containing the links to the STAC catalogs for the available sources.

Parameters
`None.` –

Returns
None.

`stacCuration.`**`createHPCSTACCatalog`**`()`

Create STAC catalog for the HPC centers.

For each of the HPC centers, create the STAC catalog containing the links to the STAC catalogs for the Apps.

Parameters
`None.` –

Returns
None.

`stacCuration.`**`createImageSrcListForSTAC`**`(`*srcCatalog*, *srcFileList*`)`

Create STAC items for the image files in the input
file list of files.

For each of the image files in the input file list, create the STAC item and save in the input catalog file.

Parameters

- **`catFile`** – Catalog File Name.
- **`srcFileList`** – List of image files.

Returns
None

`stacCuration.`**`createNetcdfSrcListForSTAC`**`(`*srcCatalog*, *srcFileList*`)`

Create STAC items for the netcdf files in the input
file list of files.

For each of the netcdf files in the input file list, create the STAC item and save in the input catalog file.

Parameters

- **`catFile`** – Catalog File Name.
- **`srcFileList`** – List of netcdf files.

Returns
None

`stacCuration.`**`createTextSrcListForSTAC`**`(`*srcCatalog*, *srcFileList*`)`

Create STAC items for the text files in the input
file list of files.

For each of the text files in the input file list, create the STAC item and save in the input catalog file.

Parameters

- **`catFile`** – Catalog File Name.
- **`srcFileList`** – List of text files.

Returns
None

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